Change Detection in Remote Sensing Images Using an Improved Transformer Architecture

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***Abstract*—To address the challenge that change detection in RS images still struggles to accurately identify and localize changed areas—mainly due to environmental complexity, illumination variations, and occlusions—this paper proposes an improved change detection algorithm called SSC (STANet-Swiftformer with CBAM). Using STANet as the baseline framework, this algorithm combines it with the Transformer-based Swiftformer model, inheriting STANet's excellent feature extraction and multi-scale analysis capabilities while introducing Swiftformer's global context awareness and efficient self-attention mechanism. Furthermore, the Convolutional Block Attention Module (CBAM), which simulates the human visual attention mechanism, is embedded to guide the network to concentrate on key areas in the images, thereby significantly enhancing the accuracy and robustness of feature representation. Through this innovative design, the SSC algorithm can effectively capture the local details of remote sensing images while comprehensively grasping the global contextual information, thereby achieving more accurate identification of changing areas. Experimental results demonstrate that the SSC algorithm achieves significant betterment in both detection accuracy and comprehensive performance.**

Keywords-Remote Sensing Images; Change Detection; Deep Learning; Convolutional Neural Network(CNN)

I. INTRODUCTION

Advancements in remote sensing technology have driven an evolution in image quality, marked by the transition from low-resolution multispectral to high-resolution hyperspectral capabilities. High image quality is pivotal in remote sensing change detection, forming the foundation for its widespread applications in urban planning[1], land cover classification[2], disaster assessment[3], and environmental monitoring[4]. However, the images contain various ground objects with different colors, shapes, and textures, resulting in complex information. They are also affected by factors such as shadows and noise, are significantly influenced by illumination angle, and the actual changed areas typically occupy a very small proportion of the entire image. All these factors pose challenges for accurately identifying changes in multi-temporal remote sensing images.

In the past decade, deep learning-based change detection algorithms have undergone rapid development. These algorithms can automatically learn and extract change features from images by training neural network models, possessing strong adaptive capabilities and higher detection accuracy. Researchers worldwide have achieved significant results in this field. Daudt et al.[5] proposed three fully convolutional models, FC-Siam-conc, FC-Siam-diff, and FC-EF, which improve change detection accuracy by adding skip connections to Siamese neural networks. Peng et al.[6] and Fang et al.[7] improved U-Net++ and introduced attention mechanisms to enhance the quality of change maps, respectively. Song et al.[8] improved the U-Net network to enhance its feature extraction capability. Several researchers have optimized model performance by fusing multi-source features, introducing attention mechanisms, and employing deep supervision[9]-[13]. Other studies combined CNN and Transformer to balance feature extraction efficiency and global modeling capability. However, the aforementioned research primarily focuses on how to better extract discriminative features and how to effectively perform feature fusion. Challenges remain in multi-scale context modeling[14] and noise suppression.

This study, therefore, introduces an integrated deep learning approach for multi-temporal remote sensing change detection. The method synergistically combines local feature extraction with global contextual understanding, significantly enhancing the accuracy of change identification and area localization. It also demonstrates excellent generalization ability, maintaining stable and reliable detection performance under various lighting conditions and complex scenes, providing a new and effective tool for remote sensing image change monitoring.

II. RESEARCH METHOD

A. Overall Network Structure

Due to its critical role in land use analysis[15], environmental monitoring, and urban planning, change detection in remote sensing images is a widely studied field. Based on current research, a new metric learning-based network model named SSC is proposed. Based on current research, a new metric learning-based network model named SSC is proposed. This model comprises three components: a feature extractor, an attention module, and a metric module. Its core lies in using the self-attention mechanism to model spatiotemporal dependencies, effectively overcoming interference caused by illumination changes and registration errors, thereby more accurately identifying

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real ground object changes. The SSC network model takes two images as input and aims to produce a label diagram of the same dimensions as the input images, which can display the change differences between the two images. First, the two images I1 and I2 are progressively passed to the Swiftformer feature extractor, obtaining corresponding two feature maps X1 and X2, where X1, X2$\in R^{(C*H*W)}$. Here, C denotes the channel depth of the feature vector, while H and W correspond to the height and width dimensions of the feature maps, respectively. Next, the attention module refines these feature maps and produces two corresponding attentive feature maps, Z1 and Z2. The pairwise pixel distance calculation between the two resized feature maps (adjusted to the input image size) is performed by the metric module, yielding a resultant distance map. In the course of training, to better reflect the changed and unchanged areas in the data, the relationship between the distance diagram and the label diagram is continuously adjusted to ensure accurate representation. Specifically, when the model identifies a changed point, it assigns a larger distance value to that point to highlight the change. Conversely, it reduces the distance between them to highlight their consistency. The model pipeline is shown in Figure 1.

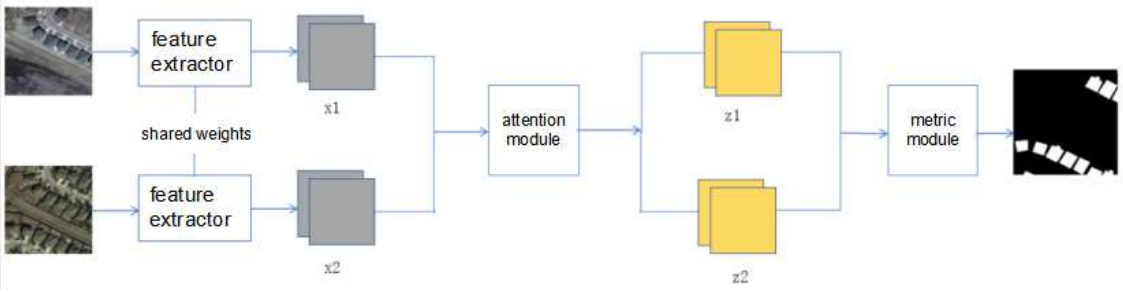


Figure 1. Model pipeline diagram

B. Feature Extractor

The feature extractor is usually composed of multi-layer convolutional neural networks or other network layers. To avoid a large number of complex multiplication operations and extract effective data features, this paper uses Swiftformer to build the feature extractor. Swiftformer employs an efficient additive attention mechanism, drawing on the attention mechanism of the Transformer, which allows the model to capture long-range dependencies[16]. This approach offers higher efficiency, lower memory consumption, and superior model generalization ability and interpretability. There are four stages, from input to output, each designed with an effective convolution. Encoder and a Swiftformer Encoder.

C. Attention Module

Leveraging the self-attention mechanism, the SSC model incorporates two dedicated modules: the Basic Spatiotemporal Attention Module (BAM) and the Pyramid Spatiotemporal Attention Module (PAM).

BAM can capture and analyze the correlation between any two positions in the spatiotemporal dimension as attention weights. This mechanism calculates the response of each position in a specific context by performing a weighted processing of all positional features in spacetime. Specifically, the feature maps X1 and X2 are stacked to form a three-dimensional feature vector X, which is input into the BAM algorithm, where X$\in R^{(C\times H\times W\times 2)}$. This results in a new feature vector Z, Z$\in R^{(C\times H\times W\times 2)}$. As Z becomes more compact, it can reflect the real data changes. Finally, the updated feature vector Z is split into two feature maps Z1 and Z2, using residual technology. The formula is as follows:

$$Z = F(X) + X \tag{1}$$

Here, Y=F(X) is the residual mapping of X.

In this way, BAM can continuously learn to adapt to the dataset, ultimately improving the prediction accuracy.

To calculate Y, it is necessary to learn a weighted sum of a set of vectors Q, K, and V in the input vector to generate the output vector. The detailed procedure is outlined as follows:

(1) The stacked feature vector X is used to generate Q and K vectors, where Q, K$\in R^{(C\times H\times W\times 2)}$. These two feature vectors are then reshaped into the key matrix K1 and the query matrix Q1, where K1, Q1 $\in R^{(C1\times N)}$, with N=H×W×2.

(2) To obtain the new feature vector V, X needs to be input into a convolutional layer (C, 1×1/1), where V$\in R^{(C1\times H\times W\times 2)}$. Then V is reshaped into the value matrix V1. Here, C1 is C/8, used to reduce the feature dimension.

(3) After obtaining K1, Q1, and V1, the similarity matrix A$\in R^{(N\times N)}$ is obtained after Softmax normalization. In the similarity matrix, any element A[i, j] represents the pairwise similarity score computed between the i-th key and the j-th query.

(4) The output matrix $Y_i \in R^{(C\times N)}$ is obtained by multiplying A and V1. It is then reshaped to get Y, and Y is multiplied by X to get Z.

In PAM, the BAM is embedded within a pyramid architecture to produce multi-scale attentive representations. PAM employs a four-branch architecture. Each branch partitions the input feature tensor into multiple sub-regions at a specific scale, and each sub-region incorporates the BAM module to calculate local attention. Finally, the output tensors of these four branches are used to generate a multi-scale attention representation. Through this hierarchical structure design, the model's capability to identify details is enhanced, allowing for more refined and flexible attention allocation, which in turn enables it to capture spatiotemporal relationships and detailed information more effectively in complex scenes. The detailed procedure is outlined as follows:

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(1) For the given dual-temporal feature maps X1, X2 $\in R^{(C\times H\times W)}$, they are stacked to form a new feature map X, where $X\in R^{(C\times H\times W\times 2)}$.

(2) Each branch divides the feature tensor equally into S×S sub-regions, and then the BAM module is added to all four branches.

(3) By applying BAM to all regions, the adjusted residual feature tensor $Y_i\in R^{(C\times H\times W\times 2)}$ is obtained.

(4) All feature tensors Y_i are concatenated and input into a convolutional layer (C, 1×1/1) to obtain the final residual feature tensor $Y\in R^{(C\times H\times W\times 2)}$. Eventually, $Y\in R^{(C\times H\times W\times 2)}$ is added to X to obtain the updated feature tensor $Z\in R^{(C\times H\times W\times 2)}$.

D. Feature Enhancement Module

The SSC model introduces the Convolutional Block Attention Module (CBAM), which simulates the human visual attention mechanism, as a feature enhancement module. This module integrates complementary channel and spatial attention mechanisms.

The primary design objective of the channel attention module is to enhance the discriminative power of feature representations across all channels. First, for the input feature map F1, the channel attention mechanism applies dual-pooling operations (global max and average) across all channels., generating vectors representing global max features and average features to capture dependencies between channels. Then, a fully connected layer processes the two generated vectors to learn per-channel attention weights, where each weight indicates the significance of the channel to the task at hand. The two vectors are subsequently combined through element-wise addition to yield the initial attention weights. To normalize these values to a [0,1] range, a Sigmoid activation function is applied, producing the final channel attention weights. Eventually, each channel of the original feature map is modulated by its corresponding attention weight, resulting in a channel-recalibrated feature map.

The goal of the spatial attention module is to emphasize the importance of different locations in the image, enhancing the model's attention more to areas crucial for the specific task. First, for the input feature image F2, max pooling and average pooling operations are performed along the channel dimension to obtain features at different contextual scales. Then, the features obtained in the previous step are stacked along the channel dimension, resulting in a new feature map with more channels and multi-scale contextual information. The feature map is then mapped into spatial attention weights via a convolutional transformation. The obtained attention weight vector is then processed through a Sigmoid activation function, normalizing the values to the [0, 1] range to generate the spatial attention weights. Finally, selective enhancement of the original feature map is performed under the guidance of spatial attention weights via element-wise multiplication. This operation emphasizes informative regions while suppressing contributions from less relevant areas, thereby enhancing model focus.

CBAM works by first taking the feature diagram F3 as input. After passing through the channel attention module, it is combined with the original feature diagram through multiplication, resulting in a refined feature diagram. Then, the output feature diagram passes through the spatial attention module to get another new feature diagram. Finally, it is multiplied by the initial image to get the final feature diagram.

In the SSC model, there is a DR class, which is a simple dimensionality reduction module. The method first compresses the feature map's channel depth using a 1×1 convolution, followed by successive processing through Batch Normalization and ReLU activation. In the DR class, a CBAM instance is added after the features of each DR module have been implemented. Through this method, after the feature map undergoes dimensionality reduction, CBAM further processes the feature map to emphasize important channels and spatial locations.

E. Metric Module

The core principle of deep metric learning lies in learning a mapping relationship through a deep neural network, which projects high-dimensional data into a low-dimensional embedding space and optimizes the similarity measurement within this space. In this feature space, the distance between similar samples is reduced, while the distance between different samples is effectively increased, enabling more accurate data analysis. In the SSC model, a contrastive loss approach is used to encourage the distance between each unchanged pixel pair to decrease and the distance between changed pixels in the embedding space to increase.

F. Loss Function

In machine learning applications, it is common to encounter datasets with skewed class distributions. Moreover, in the real world, most changed pixels account for only a small fraction of the total pixels, meaning that neural networks are easily affected by this imbalance during training. To solve this problem, we introduce a class-sensitive loss function termed Batch Balanced Contrastive Loss. For a given sample $X^{*(1)}$, $X^{*(2)}$ and M*, where $X^{*(1)}$, $X^{*(2)}\in R^{(B\times H0\times W0)}$, $M^*\in R^{(B\times H0\times W0)}$, the distance map $D^*\in R^{(B\times H0\times W0)}$ can be obtained through the SSC model, where B is the batch size, and M* is the binary label map, with 1 representing change and 0 representing no change. The batch balanced definition is as follows:

$$L\left(D^*,M^*\right)=\frac{1}{2}\frac{1}{n_u}\sum_{b,i,j}\left(1-M^*_{b,i,j}\right)D^*_{b,i,j}+\frac{1}{2}\frac{1}{n_c}\sum_{b,i,j}M^*_{b,i,j}\,\mathrm{Max}\left(0,m-D^*_{b,i,j}\right) \tag{2}$$

Where b, i, and j represent the batch number, height, and width. m is 1, nu, and nc correspond to unchanged and changed pixel pairs.

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III. EXPERIMENTAL SETUP

A. Dataset

This paper utilizes the publicly available LEVIR-CD dataset. The original dataset includes 445 pairs of training images, 64 pairs of validation images, and 128 pairs of test images. Each image is meticulously cropped to 1024×1024 to ensure the model captures sufficient detail while maintaining computational efficiency. Due to GPU resource limitations, this experiment divided the original images into non-overlapping 256×256-pixel patches, preserving the original image information while maintaining the overall spatial resolution.

B. Experimental Environment and Parameter Settings

The model implementation was carried out utilizing the PyTorch framework. The Adam optimizer was used to update the network model parameters, with the Adam momentum parameter set to 0.5, with a learning rate of 0.001 and a batch size of 8. The learning rate was decayed linearly at intervals of 50 epochs. For the loss function design, the margin parameter used to define the contrastive loss was initialized to 2.0.

C. Evaluation Metrics

To validate the efficacy of the presented approach, Precision (Pr_i), Recall (Re_i), F1-score ($F1_i$), Overall Accuracy (OA), and Mean_IOU are used as evaluation metrics for the model. The calculation formulas are:

$$Pr_i = n_{ii} / \sum_j n_{ji} \tag{3}$$

$$Re_i = n_{ii} / \sum_j n_{ij} \tag{4}$$

$$F1_i = 2 Pr_i Re_i / (Pr_i + Re_i) \tag{5}$$

Where n_{ii} represents the number of pixels whose true category is i and are predicted as true category j.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Ablation Study

To comprehensively analyze the specific contributions of the two new modules to the model performance, ablation experiments were specifically conducted for these two core modules. The specific metric performance is shown in Table I.

TABLE I. COMPARISON RESULTS OF ABLATION EXPERIMENTS

	Swiftformer	Attention Mechanism	Pre	Rec	F1	OA	Mean_Iou
Base			60.1	91.7	72.6	96.5	0.766 0:0.963 1:0.570
Base	√		68.5	92.5	78.5	97.5	0.811 0:0.973 1:0.649
PAM			75.3	94.4	83.8	98.1	0.850 0:0.980 1: 0.721
SSC	√	√	83.4	92.2	87.6	98.7	0.883 0:0.986 1:0.779

The red font indicates the optimal results, while the blue font represents the suboptimal results.

According to Table I, the SSC algorithm achieved the best results across multiple key performance indicators, demonstrating its high efficiency and remarkable capability in detecting all positive samples. For the baseline model (Base), the integration of the Swiftformer module led to comprehensive performance improvements. This outcome not only confirms the effectiveness of the Swiftformer module as an efficient feature extractor but also validates its suitability and importance in the proposed algorithm.

Furthermore, the PAM model was optimized and refined in this experiment. Originally, the feature extractor of the PAM model was based on ResNet18. In this study, the Swiftformer module was used in place of the Original module, and the CBAM module was introduced for feature enhancement. The revised model achieved strong results across several key performance metrics, even ranking first in the recall indicator, which fully demonstrates the effectiveness of incorporating the CBAM module.

In summary, the algorithm in this paper demonstrates superior performance in multiple aspects. Especially after replacing the feature extractor with the Swiftformer module and introducing the CBAM module, the situations of connection between boundaries and missed or false detections by the network were improved, and the model's performance was comprehensively enhanced. Figure 2 displays selected visualization results from the experiment.

From left to right: T1, T2, Ground Truth, Base, Base+Swiftformer, PAM, SSC

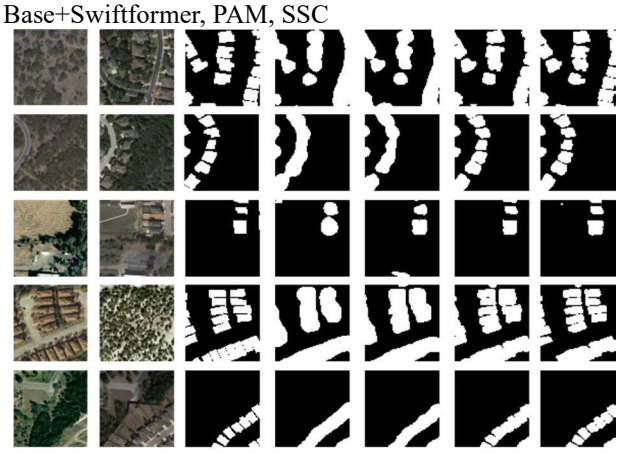


Figure 2. Ablation experiment results.

As clearly shown in the first, second, fourth, and fifth groups of Figure 2, replacing the feature extractor in the Base model with Swiftformer enables more accurate identification of subtle differences between boundaries. This results in more distinct transitions along edges and effectively prevents large-scale adhesion. To boost the model's performance, the CBAM module was incorporated on top of Swiftformer. This allows the model to focus more on features sensitive to densely changed areas. Experimental results show that the improved model preserves relatively complete boundaries for both large and small change areas. This can be explained by the

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synergistic effect arising from the collaboration between Swiftformer and CBAM, which enables the model to capture fine-grained boundary variations more accurately and enhances edge information to maintain the integrity of changed regions. Moreover, due to the model's increased attention to critical features, instances of missed and false detections are correspondingly reduced.

In summary, by adopting Swiftformer as the feature extractor and integrating the CBAM module, the SSC algorithm demonstrates significant performance improvements in detecting densely changed regions. This not only confirms the effectiveness of Swiftformer in feature extraction but also highlights the benefits of introducing the CBAM attention mechanism in reducing missed and false detections.

B. Comparative Experiment

To better demonstrate the effectiveness of the algorithm, a comparative experiment was conducted under the same experimental conditions between the SSC algorithm and two other remote sensing image processing algorithms, BIT-CD and STANet, to evaluate the performance of the new algorithm. The specific metric performance is shown in Table II.

TABLE II. COMPARISON RESULTS OF COMPARATIVE EXPERIMENTS

	OA	Mean_IoU	Pre	Rec	F1
STANet	0.982	0.8560:0.9811:0.730	0.767	0.938	0.844
BIT	0.973	0.8600:0.9721:0.580	0.740	0.729	0.734
SSC	0.987	0.8830:0.9861:0.779	0.834	0.922	0.876

The red font indicates the optimal results, while the blue font represents the suboptimal results.

According to Table II, it can be seen that the SSC algorithm exhibits superior comprehensive performance when processing the LEVIR-CD dataset, particularly in key metrics such as Precision and F1-score, which further verifies the strong capability of this algorithm in the domain of change detection in RS images. Figure 3 displays selected visualization results from the experiment.

From left to right: T1, T2, Ground Truth, STANet, BIT, SSC

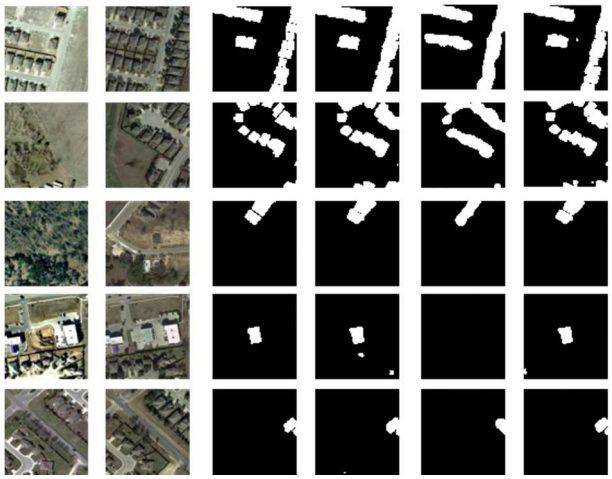


Figure 3. Comparative experiment results.

As shown in Figure 3, the SSC algorithm exhibits significant advantages in detecting changed areas, with the most notable effect.

For those small area changes that are difficult to detect in the image, as shown in the third, fourth, and fifth groups of experiments, SSC can accurately locate and identify these subtly changed parts. At the same time, although the BIT algorithm and the STANet algorithm can also effectively identify some changed areas, they still have shortcomings in some aspects. For example, in the fourth group of experiments, the STANet algorithm incorrectly identified the shadow area as a changed area, while the BIT algorithm failed to detect any changed area in the same group of experiments. This further highlights the advantage of the SSC algorithm in detecting small area changes. In the first and second groups of experiments, for densely changed areas, the SSC algorithm also demonstrated excellent performance, accurately identifying the changed areas and avoiding large-scale adhesion. This advantage is more pronounced when detecting large-scale changed areas, as the detection results of the SSC algorithm are more comprehensive and accurate.

In summary, whether in detecting small area changes or large-scale changed areas, the SSC algorithm demonstrates excellent performance.

V. CONCLUSION

Deep learning algorithms, with their powerful data-driven capability and feature learning ability, can automatically learn and identify key features in images while filtering out noise and interference introduced by natural factors. Through the learning and training of deep neural networks, the model can more accurately identify the areas where changes have actually occurred, improving the precision and trustworthiness of change detection.

This paper conducted research on Transformer-based RS image change detection methods and proposed the SSC algorithm. This algorithm fully utilizes the advantages of the Transformer in sequence data processing, performing efficient and accurate feature extraction and change detection for remote sensing images through mechanisms such as self-attention and positional encoding. Through extensive experimental verification and comparative analysis, the validity of the SSC algorithm in RS image change detection tasks is proven. Future research will aim to improve and expand the identification of specific types of changes detected by this algorithm, as well as further optimize it.

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